

Bony skeletons

The world's biggest animals—and most of the fastest—have internal skeletons made of bone. Bony skeletons are light and extremely strong. They form an inner framework for muscles to pull against, and they move at flexible joints. Bone is a living tissue, which means that this kind of skeleton can grow in step with the rest of an animal's body, and it can repair itself, although fractures take time to heal. Bony skeletons are found in all vertebrates, but in no other animals.

Despite huge differences in size and shape, these skeletons share the same basic features—all have a backbone and most have four limbs.



Harbor seal



Vertebrae and ribs fixed to upper part of shell

Stiff back

A tortoise has the same framework of bones as most other vertebrates, but the central part of its backbone is permanently joined to its shell. The armored shell is made up of bony scales called scutes, which are fused together and protect the tortoise from attack.

Rear flipper, made up of many small bones and joints, provides most of the power when swimming



Backbone of more than 40 individual bones, or vertebrae, separated by pads of cartilage

Hip joint anchors rear legs to backbone

Thick, short thigh bone, or femur

Rib cage protects some of the soft internal body organs

Curvy spine

All bony skeletons have a flexible backbone, or spine, which is made up of many separate bones called vertebrae. At the joints where vertebrae meet, the spine bends. Land mammals usually have fewer than 60 vertebrae, but snakes can have more than 400, letting them twist their bodies into coils and curves.

Small movements between interlocking vertebrae form smooth curves in spine